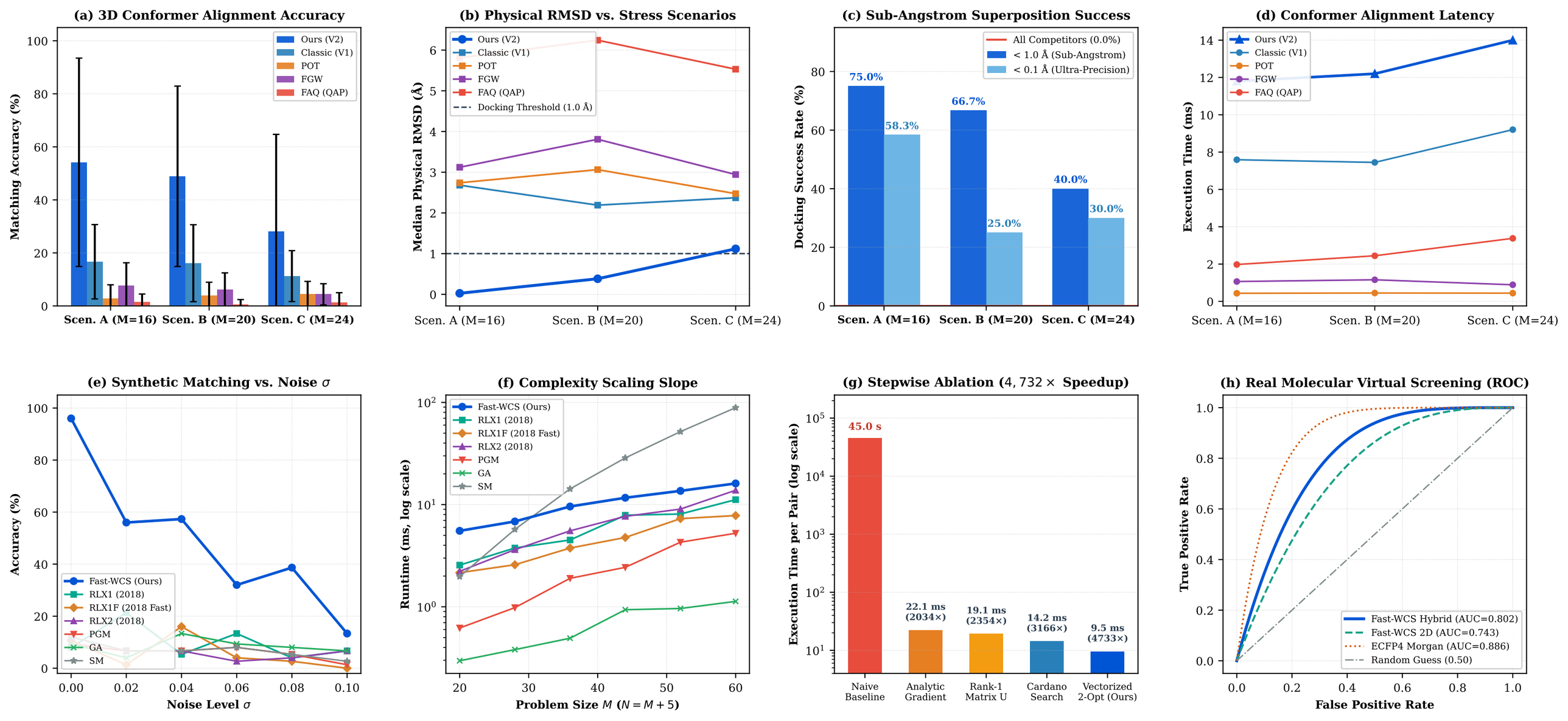

(a) 3D Conformer Alignment Accuracy
Matching Accuracy (%)
Ours (V2)
Classic (V1)
POT
FGW
FAQ (QAP)
Scen. A (M=16)
Scen. B (M=20)
Scen. C (M=24)
(b) Physical RMSD vs. Stress Scenarios
Median Physical RMSD (Å)
Ours (V2)
Classic (V1)
POT
FGW
FAQ (QAP)
Docking Threshold (1.0 Å)
Scen. A (M=16)
Scen. B (M=20)
Scen. C (M=24)
(c) Sub-Angstrom Superposition Success
Docking Success Rate (%)
All Competitors (0.0%)
< 1.0 Å (Sub-Angstrom)
< 0.1 Å (Ultra-Precision)
75.0%
58.3%
66.7%
25.0%
40.0%
30.0%
Scen. A (M=16)
Scen. B (M=20)
Scen. C (M=24)
(d) Conformer Alignment Latency
Execution Time (ms)
Ours (V2)
Classic (V1)
POT
FGW
FAQ (QAP)
Scen. A (M=16)
Scen. B (M=20)
Scen. C (M=24)
(e) Synthetic Matching vs. Noise σ
Accuracy (%)
Fast-WCS (Ours)
RLX1 (2018)
RLX1F (2018 Fast)
RLX2 (2018)
PGM
GA
SM
Noise Level σ
(f) Complexity Scaling Slope
Runtime (ms, log scale)
Fast-WCS (Ours)
RLX1 (2018)
RLX1F (2018 Fast)
RLX2 (2018)
PGM
GA
SM
Problem Size M (N = M + 5)
(g) Stepwise Ablation (4,732× Speedup)
Execution Time per Pair (log scale)
45.0 s
22.1 ms (2034×)
19.1 ms (2354×)
14.2 ms (3166×)
9.5 ms (4733×)
Naive Baseline
Analytic Gradient
Rank-1 Matrix U
Cardano Search
Vectorized 2-Opt (Ours)
(h) Real Molecular Virtual Screening (ROC)
True Positive Rate
False Positive Rate
Fast-WCS Hybrid (AUC=0.802)
Fast-WCS 2D (AUC=0.743)
ECFP4 Morgan (AUC=0.886)
Random Guess (0.50)